Chapter 6B

Diabetes Mellitus

Diagnosis, Treatment, DKA/HHS, and Chronic Complications

How to use this chapter

This chapter is an original Medvale study chapter built from uploaded coverage signals and current guideline cross-checks. It is designed for learners and for future graph-RAG extraction: concepts are linked by mechanisms, diagnostic thresholds, treatment choices, complications, and clinical traps.

Source coverage signal: the uploaded endocrine section covers diabetes classification, diagnosis, outpatient management, noninsulin and insulin therapy, chronic complications, diabetic nephropathy/retinopathy/neuropathy, diabetic foot, DKA, and HHS. This PDF uses that material only as a topic checklist and does not reproduce its prose or figures.

Guideline cross-checks: ADA Standards of Care in Diabetes 2026; ADA/EASD pharmacologic principles; 2024 consensus report on hyperglycemic crises in adults; Endocrine Society inpatient hyperglycemia guidance.

Chapter roadmap

Section	Core questions answered
1. Why diabetes matters clinically	Why hyperglycemia injures vessels, nerves, kidney, retina, myocardium, immune defenses, and acute metabolism.
2. Illness scripts and pathophysiology	How to distinguish type 1 diabetes, type 2 diabetes, LADA, secondary diabetes, and stress hyperglycemia.
3. Diagnosis and screening	How to use A1c, fasting glucose, random glucose, 2-hour OGTT, repeat testing, and prediabetes ranges.
4. Outpatient management	How lifestyle, metformin, GLP-1 receptor agonists, SGLT2 inhibitors, insulin, and other agents fit together.
5. Insulin and inpatient diabetes	How basal-bolus thinking differs from sliding scale alone, and how to prevent hypoglycemia.
6. Chronic complications	Macrovascular disease, nephropathy, retinopathy, neuropathy, foot ulcers, infection risk, and monitoring.
7. DKA and HHS	How to recognize and treat hyperglycemic crises using fluids, insulin, potassium, and careful monitoring.
8. Practice cases and Graph-RAG map	Integrated exam-style cases and extraction-ready nodes/edges.

Core schema

The most important framework: diabetes is not just a glucose number. It is a syndrome of disordered fuel partitioning. Insulin deficiency produces catabolism and ketosis; insulin resistance produces chronic hyperglycemia, hyperinsulinemia early, vascular inflammation, and beta-cell exhaustion.

1. Why this matters clinically

Diabetes mellitus is a chronic disorder of glucose homeostasis in which the body cannot appropriately move energy from the bloodstream into insulin-sensitive tissues or cannot restrain hepatic glucose production. The immediate laboratory abnormality is hyperglycemia, but the clinical disease is broader: osmotic diuresis produces dehydration and electrolyte loss, cellular underuse of glucose produces weight loss and fatigue, and long-term glycation and oxidative stress injure endothelium, nerves, retina, kidney glomeruli, and immune function.

Clinically, diabetes is one of the few internal medicine diagnoses that can appear as a screening abnormality, a chronic vascular disease, an infection risk state, a renal disease accelerator, a neuropathy syndrome, a pregnancy issue, or an acute ICU emergency. A patient may first present with polyuria and polydipsia, blurred vision, recurrent candidiasis, poor wound healing, myocardial infarction, albuminuria, or diabetic ketoacidosis. The physician's task is to decide whether this is absolute insulin deficiency, relative insulin deficiency with resistance, medication-induced hyperglycemia, pancreatic endocrine failure, or a stress response superimposed on preexisting disease.

The high-yield numerical anchors are stable: diabetes can be diagnosed with A1c >=6.5%, fasting plasma glucose >=126 mg/dL, 2-hour plasma glucose >=200 mg/dL after a 75-g oral glucose tolerance test, or random plasma glucose >=200 mg/dL with classic symptoms. Prediabetes includes A1c 5.7-6.4%, fasting glucose 100-125 mg/dL, or 2-hour OGTT glucose 140-199 mg/dL. These numbers matter because they define when chronic hyperglycemia is likely enough to predict microvascular risk, but they do not replace clinical judgment in anemia, pregnancy, hemoglobin variants, recent transfusion, kidney failure, or acute illness.

Clinical progression: from metabolic risk to organ injury

Insulin resistance or → Hyperglycemia and dyslipidemia → Endothelial and neural injury → Nephropathy, retinopathy, → MI, stroke, PAD, infection

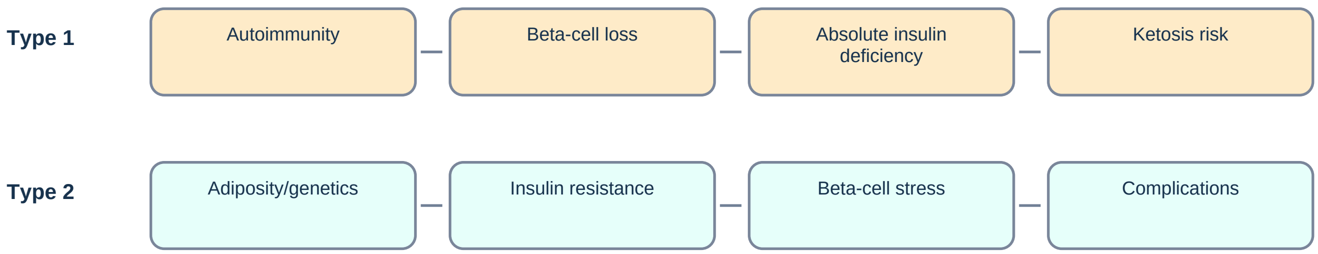
The uploaded source section emphasizes the same broad span: diagnostic criteria, therapy, monitoring, DKA/HHS, and chronic microvascular and macrovascular complications. The page images show diabetes beginning in the pancreas section and then moving from classification to diagnosis, treatment, complications, and emergencies.

2. Core illness scripts

Syndrome	Typical patient pattern	Pathophysiology	Exam/lab clues	Immediate management implication
Type 1 diabetes	Often youth or lean adult, but can occur at any age; rapid symptoms over days-weeks	Autoimmune beta-cell destruction causing absolute insulin deficiency	Ketosis, weight loss, low/absent C-peptide, autoantibodies may be present	Requires lifelong insulin; DKA risk if insulin is stopped
Type 2 diabetes	Often adult with central adiposity, family history, metabolic syndrome; increasingly seen in youth	Insulin resistance plus progressive beta-cell dysfunction	Acanthosis nigricans, hypertension, hypertriglyceridemia, albuminuria	Lifestyle plus medication chosen by A1c, ASCVD/HF/CKD, weight, cost, hypoglycemia risk
LADA	Adult who looks like type 2 at first but progresses to insulin requirement faster	Autoimmune beta-cell loss with slower tempo than classic type 1	GAD or other antibodies; declining C-peptide	Do not delay insulin when catabolic features or ketosis appear
Pancreatogenic/secondary diabetes	History of chronic pancreatitis, pancreatic surgery, cystic fibrosis, hemochromatosis, steroids, antipsychotics, transplant meds	Loss of endocrine pancreas or medication-induced resistance/secretion defects	Exocrine pancreatic symptoms or medication exposure	Treat glucose but also remove cause or manage pancreatic disease

Syndrome	Typical patient pattern	Pathophysiology	Exam/lab clues	Immediate management implication
Stress hyperglycemia	Acute MI, sepsis, trauma, surgery, glucocorticoids, tube feeds	Counterregulatory hormones and inflammatory insulin resistance	May normalize after illness; A1c helps identify chronicity	Inpatient insulin protocols; reassess after recovery

Original concept map: insulin deficiency vs insulin resistance



Classification trap

Exam trap: the labels type 1 and type 2 are not assigned by age alone. Adults can have autoimmune diabetes; adolescents can have type 2 diabetes. The emergency clue is catabolism or ketosis, which tells you insulin deficiency is clinically important right now.

3. Pathophysiology: the metabolic logic

Insulin is an anabolic signal. It promotes glucose uptake in skeletal muscle and adipose tissue, suppresses hepatic gluconeogenesis and glycogenolysis, promotes glycogen and lipid storage, and restrains lipolysis and ketogenesis. Glucagon, catecholamines, cortisol, and growth hormone oppose insulin during fasting, illness, or stress. Diabetes develops when insulin signal is insufficient for the body's metabolic demand.

In type 1 diabetes, autoimmune destruction of pancreatic beta cells gradually reduces insulin secretory reserve. Symptoms usually become obvious only after most beta-cell function is lost. Without enough insulin, adipose tissue releases free fatty acids, the liver converts them to ketone bodies, and an anion gap metabolic acidosis develops. This is why DKA is a signature presentation of severe insulin deficiency.

In type 2 diabetes, insulin resistance commonly starts in skeletal muscle, adipose tissue, and liver. Adiposity increases circulating free fatty acids and inflammatory adipokines, impairing insulin signaling. The pancreas initially compensates with higher insulin output, but chronic demand and glucotoxicity eventually exhaust beta cells. Hyperglycemia then becomes persistent and worsens vascular injury.

Microvascular complications correlate with chronic glycemia because capillaries, glomeruli, and small nerves are highly exposed to glucose-mediated damage. Macrovascular disease is driven by overlapping processes: dyslipidemia, hypertension, endothelial dysfunction, platelet activation, inflammation, kidney disease, and smoking. This is why diabetes care is not complete unless blood pressure, lipids, smoking, kidney protection, and foot/eye screening are addressed.

Mechanism	Where it shows up clinically	Test or clue
Osmotic diuresis from filtered glucose	Polyuria, nocturia, dehydration, polydipsia, hypernatremia in HHS	Very high serum glucose; urine glucose; prerenal azotemia
Protein glycation and oxidative stress	Retinopathy, nephropathy, neuropathy	A1c trend; urine albumin-creatinine ratio; dilated eye exam
Endothelial dysfunction and dyslipidemia	CAD, stroke, PAD, erectile dysfunction	ASCVD risk, LDL, triglycerides, ABI if PAD symptoms

Mechanism	Where it shows up clinically	Test or clue
Absolute insulin deficiency	Ketosis, weight loss, DKA	Beta-hydroxybutyrate, anion gap acidosis, low C-peptide
Autonomic nerve injury	Gastroparesis, orthostasis, bladder dysfunction, hypoglycemia unawareness	Symptoms; gastric emptying if unclear; orthostatic vitals

4. Diagnostic approach and screening

The diagnosis of diabetes should be made with a validated laboratory test. In a stable, asymptomatic patient, abnormal results are generally confirmed with repeat testing, either the same test repeated or a different diagnostic test. In a symptomatic patient with classic hyperglycemic symptoms and a random plasma glucose >=200 mg/dL, immediate diagnosis and treatment are appropriate. Always interpret A1c in context: it may be misleading when red blood cell lifespan is altered, such as hemolysis, recent transfusion, severe anemia, some hemoglobin variants, pregnancy, advanced kidney disease, or erythropoietin therapy.

State	A1c	Fasting plasma glucose	2-hour OGTT	Random plasma glucose
Normal	<5.7%	<100 mg/dL	<140 mg/dL	Not used alone
Prediabetes	5.7-6.4%	100-125 mg/dL	140-199 mg/dL	Not used alone
Diabetes	>=6.5%	>=126 mg/dL	>=200 mg/dL	>=200 mg/dL with classic symptoms or crisis

Screening logic

Screening is typically targeted to adults with age or risk factors, and earlier testing is appropriate in patients with overweight/obesity plus risk factors such as family history, high-risk ancestry, hypertension, dyslipidemia, prior gestational diabetes, polycystic ovary syndrome, cardiovascular disease, or sedentary lifestyle.

After diagnosis, the first pass should classify severity and risk rather than merely naming diabetes. Ask whether the patient is catabolic, ketotic, dehydrated, pregnant, acutely ill, or likely to have type 1 diabetes. Initial studies often include A1c if not already known, basic metabolic panel, creatinine/eGFR, urine albumin-creatinine ratio, lipid panel, liver-associated tests when clinically relevant, and ketones if symptoms suggest insulin deficiency or crisis.

Diagnostic workflow

Confirm hyperglycemia → Assess symptoms and crisis → Classify likely phenotype → Screen for complications → Select individualized

5. Outpatient management: treat risk, not just glucose

Initial management begins with education and shared decision-making. Patients need to understand the meaning of A1c, home glucose or continuous glucose monitoring when used, hypoglycemia recognition, sick-day rules, medication adherence, foot care, vaccinations, and when to seek urgent care. Lifestyle therapy is not a punishment; it is the baseline treatment that makes medications safer and more effective.

For many nonpregnant adults, a general A1c goal below 7% is a common starting target, but goals should be individualized. Tighter goals may fit younger patients with low hypoglycemia risk and long life expectancy. Less stringent goals may be safer in frailty, limited life expectancy, advanced comorbidity, recurrent severe hypoglycemia, or inability to safely manage complex regimens.

Medication choice in type 2 diabetes is now phenotype-driven. Metformin remains useful, inexpensive, weight-neutral to modest weight-loss, and low hypoglycemia risk, but the presence of established atherosclerotic cardiovascular disease, heart failure, chronic kidney disease, obesity, or high cardiorenal risk can justify early GLP-1 receptor agonist and/or SGLT2 inhibitor therapy independent of metformin sequencing. Insulin is appropriate at diagnosis when there are catabolic symptoms, suspected type 1 diabetes, ketosis, very high glucose, or severe symptomatic hyperglycemia.

Class	Main glucose action	Useful when	Major cautions / adverse effects
Metformin	Decreases hepatic glucose production and improves insulin sensitivity	Most type 2 diabetes patients unless contraindicated	GI upset; lactic acidosis risk in severe renal/hepatic/hypoxic states; hold around select contrast/acute illness contexts
GLP-1 receptor agonists	Increase glucose-dependent insulin, decrease glucagon, slow gastric emptying, increase satiety	Weight loss, ASCVD benefit for selected agents, high A1c with obesity	GI effects; gallbladder disease; avoid in medullary thyroid carcinoma/MEN2 history; consider gastroparesis
SGLT2 inhibitors	Increase urinary glucose excretion	CKD or heart failure benefit for selected patients; weight/BP reduction	Genital infections, volume depletion, euglycemic DKA risk, perioperative sick-day holding
Sulfonylureas/meglitinides	Increase insulin secretion	Cost-sensitive situations needing oral potency	Hypoglycemia and weight gain; caution in older adults and CKD
DPP-4 inhibitors	Increase endogenous incretin effect	Modest A1c effect, weight neutral, low hypoglycemia risk	Generally modest efficacy; dose adjust many in CKD; avoid saxagliptin/alogliptin in some HF contexts
Thiazolidinediones	Improve insulin sensitivity via PPAR-gamma	Insulin resistance when cost matters and no HF	Weight gain, edema, HF exacerbation, fractures
Insulin	Replaces or supplements insulin directly	Type 1 diabetes, DKA/HHS, catabolic type 2, severe hyperglycemia, pregnancy-specific contexts	Hypoglycemia, weight gain, education burden

Modern treatment anchor

A medication can be a glucose drug and an organ-protection drug. In current practice, SGLT2 inhibitors and GLP-1 receptor agonists are often chosen because of kidney, heart failure, ASCVD, and weight effects, not merely because of A1c reduction.

6. Insulin and inpatient diabetes

Insulin regimens are easiest to understand by separating basal, nutritional, and correction components. Basal insulin covers fasting hepatic glucose output. Nutritional insulin covers carbohydrate intake from meals or tube feeds. Correction insulin treats unexpected hyperglycemia. Sliding-scale insulin alone is reactive and often creates alternating hyperglycemia and hypoglycemia because it does not anticipate basal or meal needs.

Insulin type	Approximate onset	Common role	Clinical notes
Rapid acting (lispro, aspart, glulisine)	10-20 min	Meal and correction insulin	Dose near meals; useful for flexible carbohydrate intake
Short acting regular	30-60 min	Meals, IV insulin protocols	Only common insulin used IV in DKA/HHS protocols
NPH	1-2 h	Intermediate basal	Peak increases hypoglycemia risk; sometimes useful with steroids
Long acting (glargine, detemir, degludec)	1-4 h	Basal insulin	Do not hold basal insulin in type 1 diabetes without replacement strategy

In the hospital, persistent glucose above 140 mg/dL is generally considered hyperglycemia. Noncritically ill patients are often treated with scheduled basal-bolus or basal-plus correction regimens rather than sliding scale alone. Critical illness, DKA, HHS, perioperative instability, and rapidly changing nutrition may require IV insulin. Insulin doses must be reduced when kidney function worsens, oral intake falls, or steroids are tapered.

Patient safety anchor

Sick-day rule: illness raises counterregulatory hormones and may increase insulin needs, but poor intake raises hypoglycemia risk. Patients on insulin need explicit instructions about glucose/ketone checks, hydration, continuing basal insulin, and when to call or seek emergency care.

7. Chronic complications of diabetes

Chronic complications are best organized by vessel size and nerve involvement. Macrovascular complications affect large arteries: coronary artery disease, cerebrovascular disease, and peripheral arterial disease. Microvascular complications affect specialized small vessels: glomeruli, retinal vessels, and vasa nervorum. Neuropathy may be distal symmetric, focal mononeuropathy, cranial neuropathy, radiculoplexus neuropathy, or autonomic neuropathy. Infection risk and diabetic foot disease arise from the convergence of immune dysfunction, ischemia, neuropathy, and impaired wound healing.

Complication	Pathogenesis	Clues	Monitoring / prevention	Management focus
Diabetic kidney disease	Glomerular hyperfiltration, albuminuria, mesangial expansion, progressive sclerosis	Urine albumin-creatinine ratio 30-300 mg/g early; eGFR decline later	Annual UACR and eGFR in most; BP control	ACEi/ARB for albuminuria with HTN; SGLT2 inhibitor if eligible; glycemic/BP control
Retinopathy	Retinal capillary leakage, ischemia, neovascularization	Often asymptomatic until macular edema or proliferative disease	Dilated retinal exams at recommended intervals	Ophthalmology; anti-VEGF, laser, surgery depending stage
Peripheral neuropathy	Length-dependent nerve injury from metabolic and microvascular damage	Stocking-glove numbness, burning pain, loss of monofilament/vibration	Foot exam, monofilament, patient inspection	Foot protection; pain control; reduce ulcer risk
Autonomic neuropathy	Autonomic nerve dysfunction	Gastroparesis, diarrhea/constipation, orthostasis, bladder dysfunction, erectile dysfunction, hypoglycemia unawareness	Symptom-directed screening	Targeted therapy; medication review; fall-risk precautions
Macrovascular disease	Accelerated atherosclerosis and thrombosis	MI, stroke/TIA, claudication, PAD, erectile dysfunction	BP, lipids, smoking, kidney status	Statin, BP control, antiplatelet when indicated, SGLT2/GLP-1 when beneficial
Diabetic foot	Neuropathy plus ischemia plus infection	Painless ulcer, callus, Charcot changes, cellulitis, osteomyelitis	Regular foot checks; shoes; podiatry for high risk	Offloading, wound care, vascular assessment, antibiotics if infected

Microvascular sequence

Chronic hyperglycemia → Capillary injury → Albumin leak or ischemia → Nephropathy / retinopathy / → Organ dysfunction

Kidney complication anchor

Microalbuminuria is now usually reported as moderately increased albuminuria. The key threshold is a urine albumin-creatinine ratio of 30-300 mg/g; values above 300 mg/g indicate severely increased albuminuria. A dipstick may be negative during early albuminuria, so use UACR rather than relying on dipstick protein.

Preventive diabetes visits should link glucose data to complications. A1c is often checked every 3 months when therapy changes or goals are unmet and every 6 months when stable. Lipids, blood pressure, kidney function, albuminuria, foot risk, eye screening, immunizations, smoking, nutrition, exercise, and medication access should be reviewed systematically.

8. Acute complications: DKA and HHS

DKA and HHS are hyperglycemic crises caused by insulin deficiency plus counterregulatory hormone excess. DKA is defined by the triad of hyperglycemia or known diabetes, ketosis, and metabolic acidosis. HHS is defined by severe hyperglycemia, hyperosmolality, profound dehydration, and absent or mild ketosis/acidosis. DKA is more common in type 1 diabetes but can occur in type 2 diabetes, especially during infection, myocardial infarction, trauma, surgery, missed insulin, or SGLT2 inhibitor exposure. HHS is classically seen in older patients with type 2 diabetes who cannot keep up with free-water losses.

Both crises are treated with the same three pillars: fluids, insulin, and potassium. Fluids restore perfusion and lower glucose by dilution and improved renal clearance. Insulin stops ketogenesis and shifts potassium into cells. Potassium management prevents fatal arrhythmias because total body potassium is depleted even when the initial serum potassium is normal or high. If potassium is dangerously low, insulin must wait until potassium is replaced.

Feature	DKA	HHS
Dominant problem	Ketosis plus anion gap metabolic acidosis	Hyperosmolality and profound dehydration
Typical glucose	Often >250 mg/dL, but can be lower in euglycemic DKA	Often >600 mg/dL
pH / bicarbonate	pH <7.3 and/or bicarbonate <18 mmol/L in classic DKA	pH usually >=7.3 and bicarbonate usually >=18 mmol/L
Ketones	Beta-hydroxybutyrate elevated	Absent or mild ketones
Mental status	Variable; worsens with severity	Altered mental status common with high osmolality
Mortality	Lower with prompt treatment	Higher than DKA because patients are older and more dehydrated
Treatment focus	Fluids, insulin, potassium, close anion gap	Fluids first, cautious insulin, potassium, gradually lower osmolality

Step	DKA/HHS treatment action	Why it matters
1. Assess severity	Vitals, mental status, volume status, infection/MI/stroke triggers, BMP, venous/arterial pH, beta-hydroxybutyrate, osmolality, ECG	Identifies shock, K+ danger, precipitant, and level of care
2. Start fluids	Isotonic crystalloid initially; adjust to sodium/osmolality and hemodynamics	Restores renal perfusion and lowers glucose/osmolality
3. Check potassium before insulin	If K+ is low, replace and delay insulin until safer; if normal, give replacement with fluids; if high, monitor closely	Insulin can rapidly drop serum K+ and cause arrhythmia
4. Give insulin	Short-acting insulin IV for moderate/severe DKA or HHS; selected mild DKA may use subcutaneous protocols	Stops ketogenesis and lowers glucose
5. Add dextrose when glucose falls	Continue insulin while adding dextrose-containing fluids when glucose approaches the target range	Allows ketone clearance without causing hypoglycemia
6. Transition safely	Overlap basal insulin before stopping IV insulin; ensure patient can eat and precipitant is treated	Prevents rebound DKA

DKA trap

DKA resolves when ketoacidosis resolves, not merely when glucose normalizes. Do not stop insulin just because glucose improved; add dextrose and continue insulin until the anion gap/ketosis has cleared and the patient is ready for subcutaneous transition.

9. Differentials, mimics, and common pitfalls

Problem	Mimic/trap	How to avoid the error
High A1c but normal glucose	Iron deficiency, altered RBC turnover, hemoglobin variant, lab discordance	Compare with SMBG/CGM or fructosamine/glycated albumin when appropriate
DKA with glucose <250	SGLT2 inhibitor-associated euglycemic DKA, pregnancy, poor intake, partial insulin treatment	Check ketones and acid-base status in symptomatic patients even if glucose is not extreme
Morning hyperglycemia	Dawn phenomenon, insufficient basal insulin, nocturnal hypoglycemia with rebound, late snack	Use overnight glucose/CGM patterns rather than guessing
Neuropathic foot ulcer	Patient denies pain, so ulcer is underestimated	Perform foot exam; loss of pain is the danger signal
Albuminuria	UTI, exercise, fever, severe HTN, transient illness can raise albumin	Repeat UACR and interpret in context
HHS treatment	Lowering osmolality too quickly	Use gradual correction with frequent sodium/osmolality monitoring
Insulin in DKA	Giving insulin when K+ is severely low	Replace K+ first when low; insulin can precipitate life-threatening hypokalemia
Sliding scale alone	Reactive insulin without basal needs	Use scheduled basal/nutritional/correction strategy when indicated

10. High-yield exam anchors

Clue	Most likely association
Polyuria + polydipsia + weight loss + ketones	Type 1 diabetes or severe insulin deficiency; evaluate for DKA
Obesity + acanthosis nigricans + hypertriglyceridemia	Insulin resistance and type 2 diabetes physiology
Fasting glucose 100-125 mg/dL	Impaired fasting glucose / prediabetes
A1c >=6.5% on confirmatory testing	Diabetes mellitus
Urine albumin-creatinine ratio 30-300 mg/g	Moderately increased albuminuria, early diabetic kidney disease risk
Sudden third nerve palsy with pupil sparing	Diabetic mononeuropathy affecting CN III ischemically
Gastroparesis, orthostatic hypotension, bladder dysfunction	Autonomic neuropathy
Painful burning feet at night	Distal symmetric diabetic peripheral neuropathy
Glucose very high + osmolality high + minimal ketones	HHS
Anion gap acidosis + beta-hydroxybutyrate	DKA or ketoacidosis syndrome
K+ initially high in DKA	Serum shift despite total body potassium depletion
SGLT2 inhibitor + nausea + acidosis + modest glucose	Consider euglycemic DKA

11. Practice questions

1. A 19-year-old with 2 weeks of polyuria, weight loss, abdominal pain, Kussmaul respirations, glucose 420 mg/dL, bicarbonate 10 mmol/L, anion gap 26, and positive beta-hydroxybutyrate presents to the ED. What is the immediate treatment framework?

Answer: DKA. Treat with isotonic fluids, potassium-guided replacement, and insulin after potassium safety is assessed. Search for a precipitant such as infection or missed insulin. Glucose improvement alone is not resolution; continue insulin with dextrose until ketoacidosis resolves.

2. A 66-year-old with type 2 diabetes has glucose 850 mg/dL, effective osmolality markedly elevated, profound dehydration, confusion, pH 7.36, bicarbonate 21 mmol/L, and only mild ketones. What is the diagnosis?

Answer: HHS. The dominant problems are hyperosmolality and dehydration with absent or mild ketosis/acidosis. Treatment emphasizes aggressive but monitored fluids, potassium monitoring, and insulin, with careful avoidance of overly rapid osmolar shifts.

3. A patient with new type 2 diabetes has A1c 8.2%, established heart failure, and eGFR that allows SGLT2 inhibitor use. Which medication class has organ-protection logic beyond A1c lowering?

Answer: An SGLT2 inhibitor is favored because selected agents reduce heart failure hospitalization and slow CKD progression in appropriate patients. Therapy should still be individualized for renal function, volume status, cost, and adverse effects.

4. A patient has fasting glucose 118 mg/dL and A1c 6.0% on two checks. What category is this and why does it matter?

Answer: Prediabetes. It identifies increased risk for type 2 diabetes and cardiovascular disease, and it is a window for lifestyle intervention, weight management, and risk-factor modification.

5. A 58-year-old with diabetes has a negative urine dipstick for protein but UACR 90 mg/g. What is the interpretation?

Answer: Moderately increased albuminuria. Dipstick may miss early albuminuria; UACR is the preferred screening test for diabetic kidney disease.

6. A patient with diabetes has burning nocturnal foot pain and decreased monofilament sensation. Pulses are present. What complication is this and what prevention issue is most urgent?

Answer: Distal symmetric peripheral neuropathy. Foot protection is urgent because loss of sensation can lead to unnoticed ulcers; pain control alone does not prevent ulcers.

7. A patient taking an SGLT2 inhibitor presents with nausea, abdominal pain, anion gap acidosis, and positive ketones, but glucose is 190 mg/dL. What diagnosis must be considered?

Answer: Euglycemic DKA. The glucose is not high enough to reassure against DKA; ketones and acid-base status define the crisis.

8. An inpatient with poor oral intake is treated only with sliding-scale insulin and develops alternating hyperglycemia and hypoglycemia. What is the conceptual error?

Answer: Sliding-scale-only therapy is reactive and does not match basal or nutritional physiology. A safer plan uses basal, nutritional, and correction components adjusted to intake, renal function, and illness severity.

12. Graph-RAG extraction map

The following map is intended for future Medvale graph-RAG ingestion. Nodes should preserve definitions and numerical thresholds; edges should preserve causal, diagnostic, treatment, and complication relationships.

Node	Class	Key attributes	High-value edges
Diabetes mellitus	Disease syndrome	Chronic hyperglycemia from insulin deficiency/resistance; diagnostic thresholds A1c/glucose/OGTT	causes -> microvascular injury; causes -> macrovascular risk; treated by -> lifestyle/medications/insulin
Type 1 diabetes	Etiology subtype	Autoimmune beta-cell destruction; absolute insulin deficiency; DKA risk	requires -> insulin; causes -> ketosis when untreated
Type 2 diabetes	Etiology subtype	Insulin resistance plus beta-cell dysfunction; obesity/genetics/age risk	associated with -> metabolic syndrome; treated by -> metformin/GLP-1/SGLT2/insulin as needed
A1c	Diagnostic/monitoring test	>=6.5 diabetes; 5.7-6.4 prediabetes; approximate 2-3 month glycemia	monitors -> chronic control; limited by -> RBC turnover disorders
DKA	Acute complication	Ketosis + metabolic acidosis + hyperglycemia/diabetes; anion gap; beta-hydroxybutyrate	treated by -> fluids/insulin/potassium; triggered by -> infection/missed insulin/SGLT2
HHS	Acute complication	Severe hyperglycemia + hyperosmolality + dehydration; minimal ketosis	treated by -> fluids/insulin/potassium; causes -> neurologic symptoms
Diabetic kidney disease	Microvascular complication	Albuminuria and eGFR decline; UACR 30-300 and >300 mg/g thresholds	prevented/slowed by -> glycemic control/BP/ACEi-ARB/SGLT2
Retinopathy	Microvascular complication	Nonproliferative and proliferative disease; macular edema; screening exam	managed by -> ophthalmology/anti-VEGF/laser
Neuropathy	Microvascular/neural complication	Distal symmetric, autonomic, mononeuropathy	causes -> foot ulcer/gastroparesis/orthostasis
Diabetic foot	Complication syndrome	Neuropathy + ischemia + infection risk	requires -> foot exam/offloading/wound care/vascular assessment

Node	Class	Key attributes	High-value edges
SGLT2 inhibitor	Medication class	Glucosuria; cardiorenal benefit; euglycemic DKA risk	treats -> T2DM/CKD/HF; adverse_edge -> genital infections/volume depletion/DKA
GLP-1 receptor agonist	Medication class	Incretin effect, weight loss, ASCVD benefit for selected agents	treats -> T2DM/obesity; adverse_edge -> GI effects/gastroparesis caution

13. References used for clinical cross-checking

• American Diabetes Association Professional Practice Committee. Standards of Care in Diabetes - 2026. Diabetes Care. 2026;49(Suppl 1).

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• Uploaded source: endocrine1.pdf, diabetes mellitus section, pages showing classification, diagnosis, management, chronic complications, DKA, and HHS.